innovate achieve lead

BITS Pilani
Pilani Campus

SURVEYING (CE F213)
(Traversing)

Traversing

Traversing –(means)- Going through

In surveying Traversing –(means) – Determining the direction and length of consecutive lines.

Surveying of an AREA means to provide horizontal and vertical controls(information) of that area.

For providing horizontal controls(information) of the area, Traversing or Triangulation is used. Traversing is used for small to moderate area.

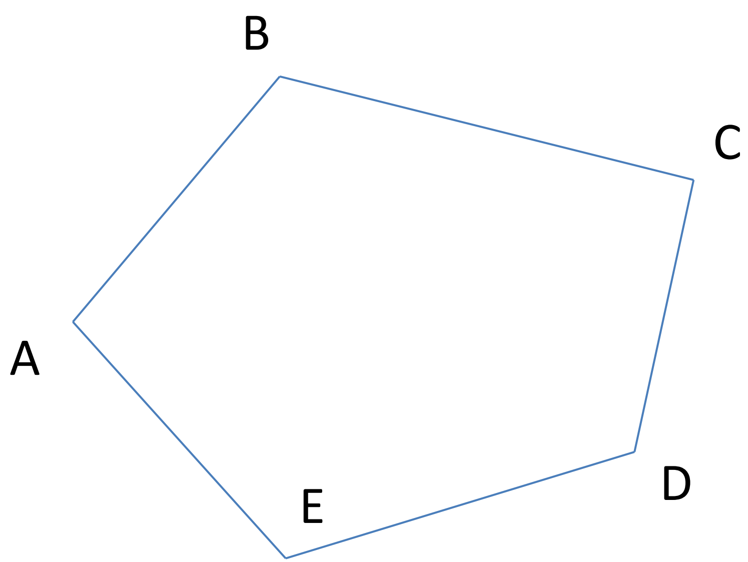
Control means- Fixing the station points accurately on the drawing sheet, because the details of the considered area is later plotted on the sheet with respect to the station points fixed in the ground.

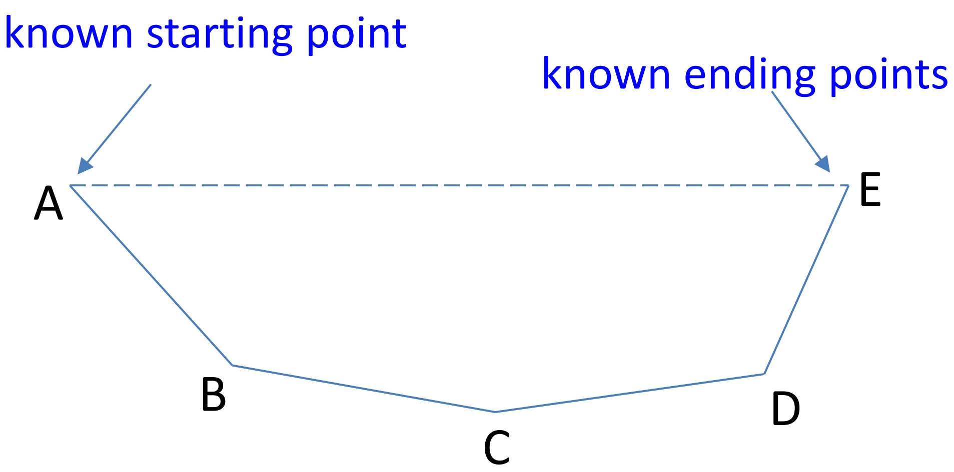
Types of Traverse

(i) Closed Traverse

(ii) Open Traverse

Closed Traverse

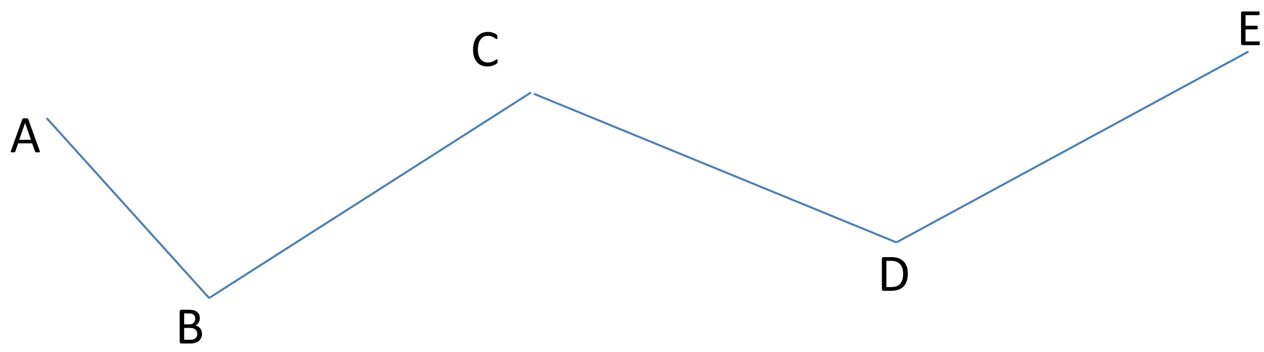




In the Closed traverse the end point returns to the starting point. A traverse whose starting point and ending points are already known in the ground is called also closed traverse.

In Open traverse the end point does not return to the starting point. The starting point and ending points are also not known previously.

Open Traverse



Open Traverse can not be checked and adjusted.

Methods of traversing

Linear measurement by Chain
Angular measurement by Compass/Theodolite

(i) Chain Traversing
(ii) Chain and Compass Traversing (Free or Loose needle method)
(iii) Chain and Theodolite Traversing (Fast needle method)
(iv) By measuring angles between lines (Theodolite traversing)
(v) Plane Table Traversing

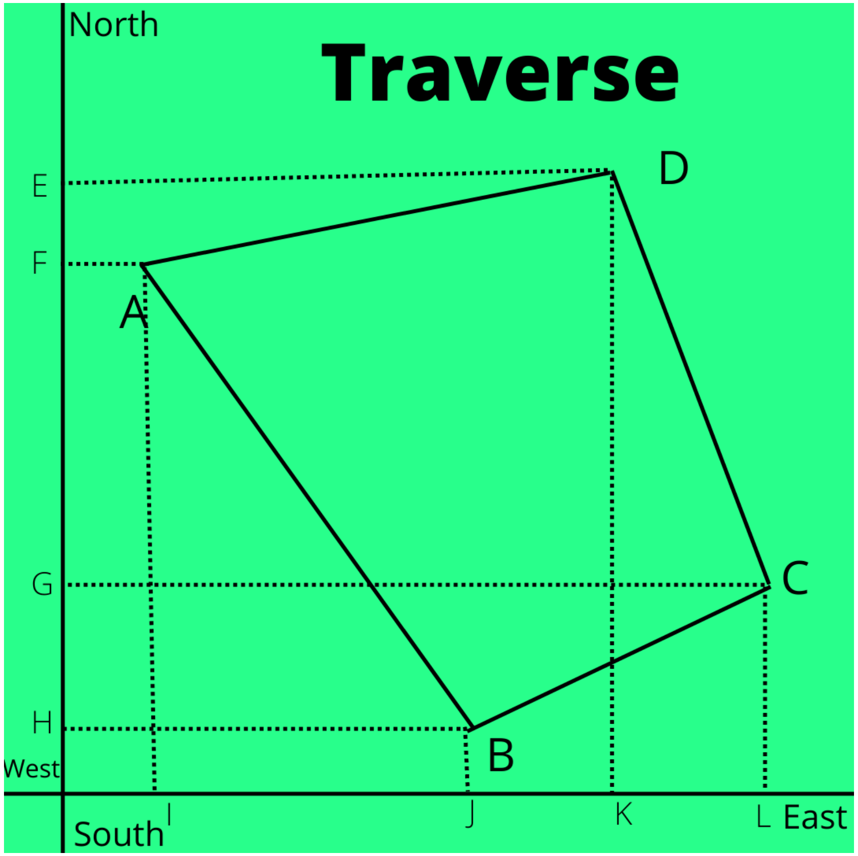
Plotting of Traversing

(i) Angle and distance method
(ii) Co-ordinate method

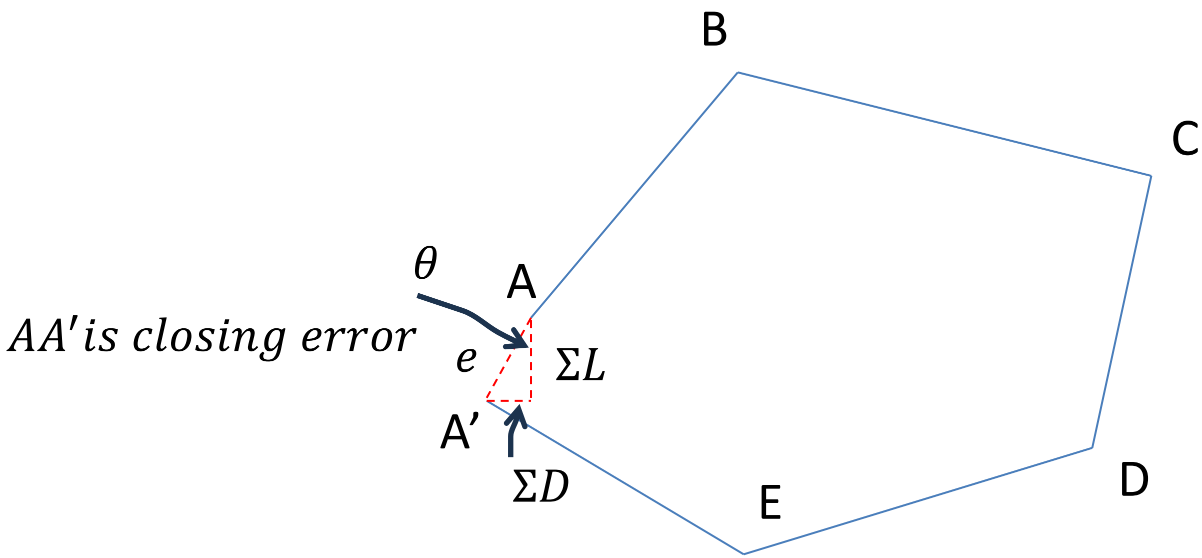
Angle and distance method

(a) By parallel meridian through each station
(b) By included angles
(c) By tangents
(d) By chords

Co-ordinate method



Closing error in a closed traverse:

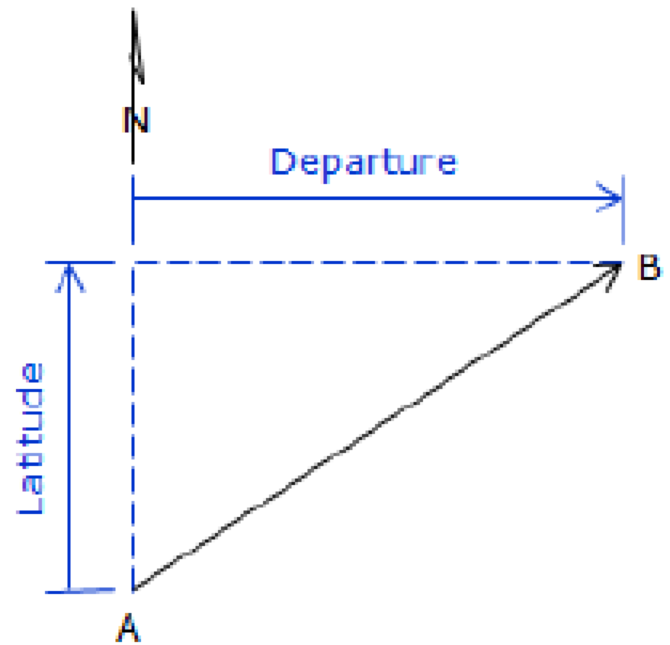


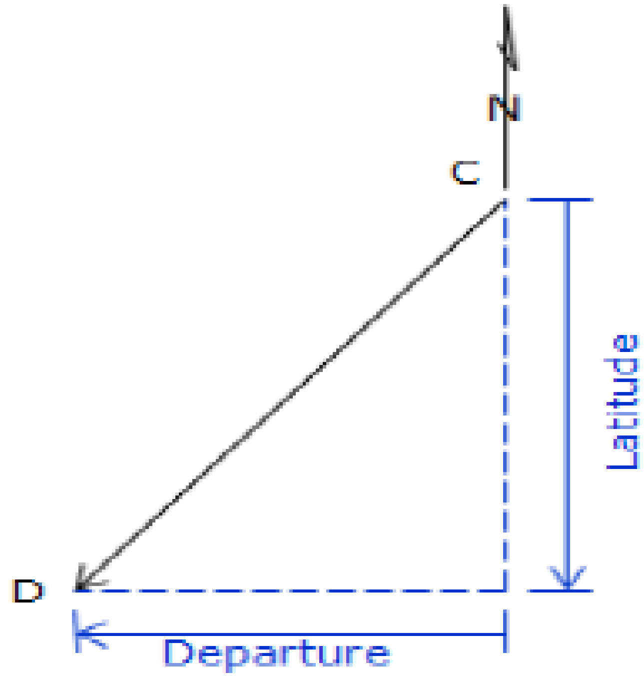
$$e = closing\ error = AA' = \sqrt{(\Sigma L)^2 + (\Sigma D)^2}$$

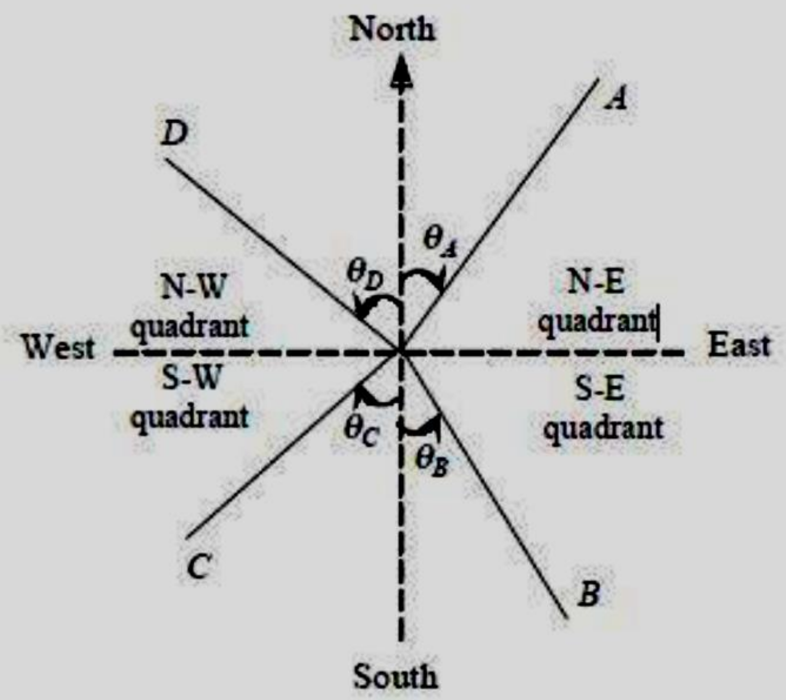
$$\text{direction of closing error is, } \tan\theta = \frac{\Sigma D}{\Sigma L}$$

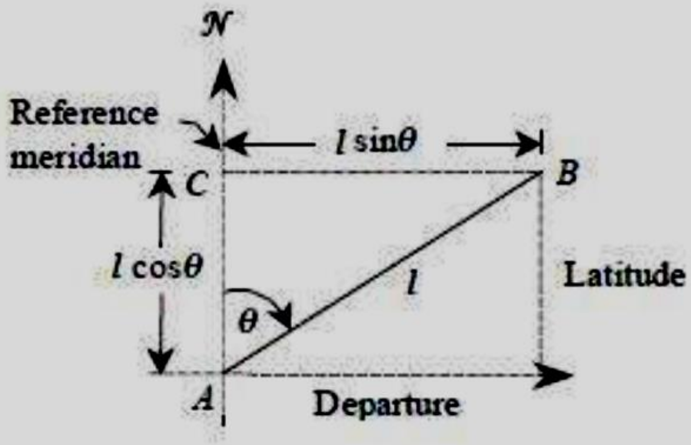
$$\text{precision of closing error} = \frac{e}{perimeter\ of\ the\ traverse}$$

Latitude and Departure

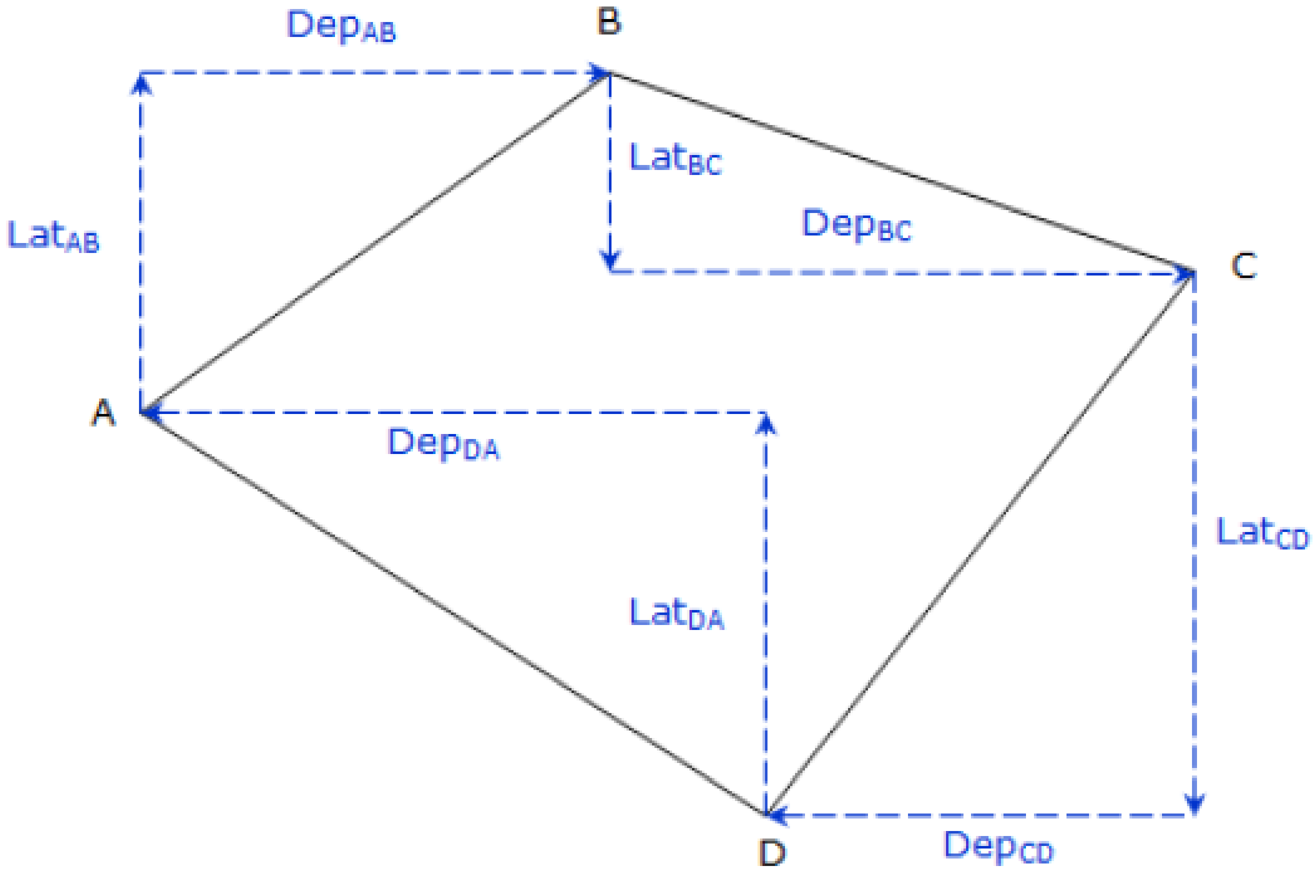








What will be nature of Latitude and Departure in different quadrant ?



$$\Sigma D = 0 \; and \;\; \Sigma L = 0 \;\; , in\; a\; closed\; perfect\; traverse$$

Balancing of Traverse

If closing error exists, then the traverse can be balanced by the following methods,

(i) Bowditch rule
(ii) Transit rule
(iii) Graphical method
(iv) Axis correction method

Bowditch rule : The error in departure and latitude will be distributed to the departure and latitude of **each side** as follows:

$$E_D = \Sigma D \frac{l}{\Sigma l}$$

$$E_L = \Sigma L \frac{l}{\Sigma l}$$

Where,

E_D – Error to the deparature of the side

E_L – Error to the latitude of the side

l – length of the side

Σl – sum of lengths of all sides – perimeter of the traverse

ΣD – Total error in departure

ΣL – Total error in latitude

Transit rule : The error in departure and latitude will be distributed to the departure and latitude of **each side** as follows:

$$E_D = \Sigma D \, \frac{d}{D}$$

$$E_L = \Sigma L \, \frac{l}{L}$$

Where,

$E_D -$ Error to the departure of the side

$E_L -$ Error to the latitude of the side

$d -$ depatture of the side (only mod value, absolute value or positive value)

$l -$ latidude of the side (only mod value, absolute value or positive value)

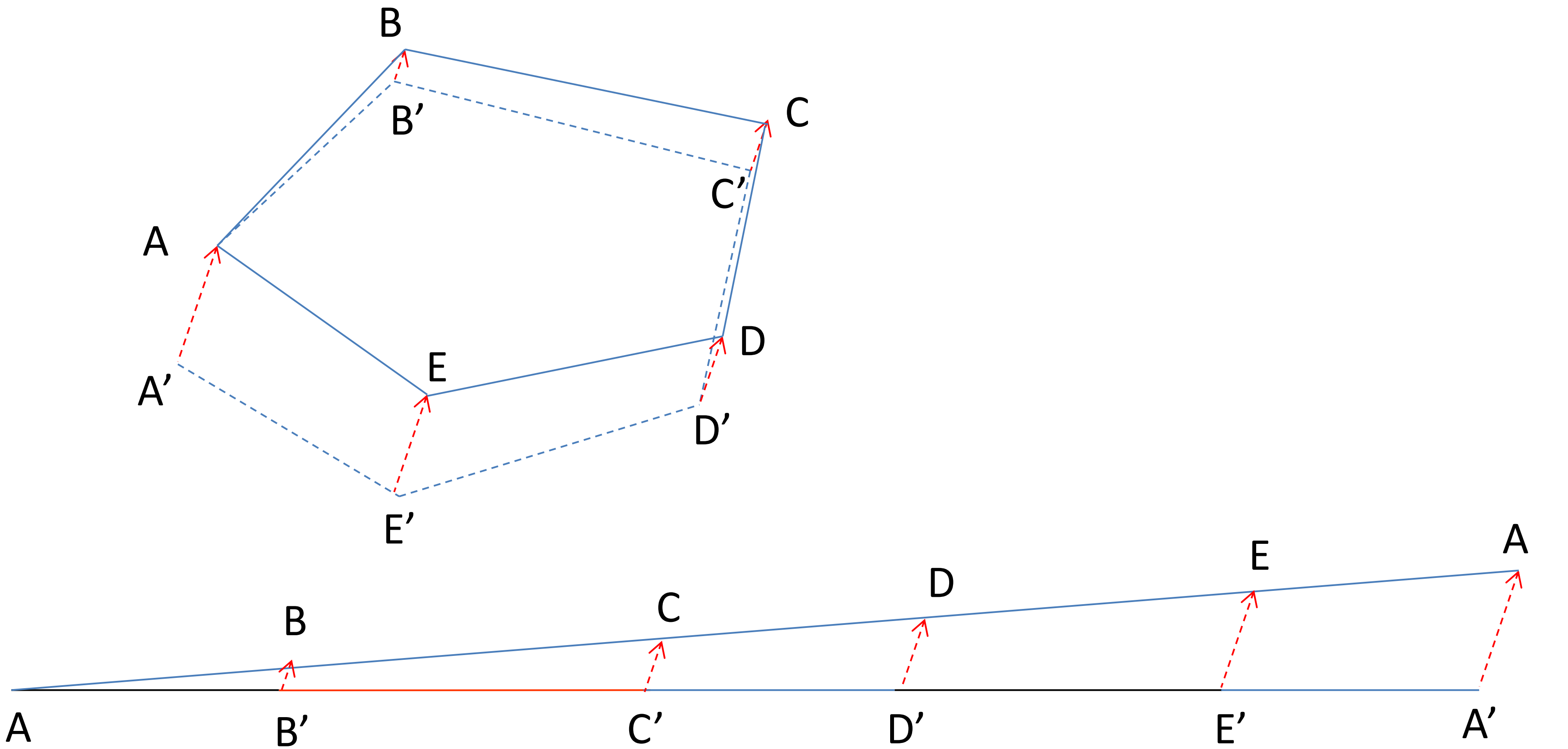
$D -$ Arithmatic sum of all departures(i. e. taking mod values only)

$L -$ Arithmatic sum of all latitudes(i. e. taking mod values only)

$\Sigma D -$ Total error in departure(Algebaric sum)

$\Sigma L -$ Total error in latitude(Algebaric sum)

Graphical method



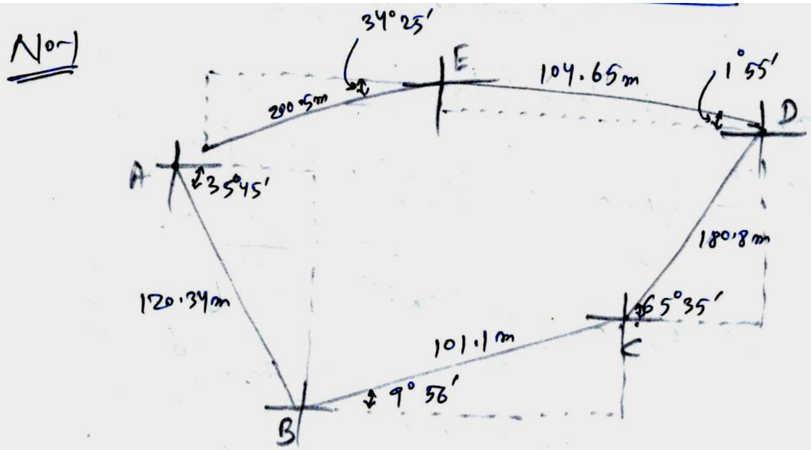
Q1: Calculate the latitudes, departures and closing error of the following traverse. Adjust the traverse by Bowditch Rule.

Line	Length(m)	FB of lines (W.C.B)
AB	89.31	45^{o} 10’
BC	219.76	72^{o} 05’
CD	151.18	161^{o} 52’
DE	159.10	228^{o} 43’
EA	232.26	300^{0} 42’

Solution

Q2: Calculate the latitudes, departures and closing error of the following traverse. Adjust the traverse by Transit Rule.

Line	Length(m)	FB of lines (W.C.B)
AB	120.34	125^{o} 45’
BC	100.1	80^{o} 4’
CD	180.8	24^{o} 25’
DE	104.65	271^{o} 55’
EA	200.5	235^{0} 35’



Line AB

Latitude $= -120.34 \sin(35°45') = -70.308$ m

Departure $= 120.34 \cos(35°45') = 97.664$ m

Line BC.

Latitude $= 101.1 \sin(9°56') = 17.267$ m

Departure $= 101.1 \cos(9°56') = 98.599$ m

Line CD

Latitude $= 180.8 \sin(65°35') = 164.629$ m

Departure $= 180.8 \cos(65°35') = 74.737$ m

Line DE

Latitude $= 104.65 \sin(1°55') = 3.5$ m

Departure $= -104.65 \cos(1°55') = -104.59$ m

Line EA

Latitude $= -200.5 \sin(34°25') = -113.324$ m

Departure $= -200.5 \cos(34°25') = -165.402$ m

5

Algebraic Sum.

$\Sigma L = 1.764$ m

$\Sigma D = 1.007$ m

Closing error

$= \sqrt{(\Sigma L)^2 + (\Sigma D)^2}$

$= 2.031$ m

2

To adjust By Transit Rule. (Arithmetic Sum) (2)

L = Sum of all latitudes (taking all +ve) = 369.028 m

D = Sum of all departures (taking all +ve) = 540.993 m

Error in Latitude and departure

(3)

Line AB

E_L = error in latitude $= \Sigma L \times \frac{l_{AB}}{L}$ (All +ve)

$= \frac{1.764}{369.028} \times l_{AB} = 0.33$

0.00478

E_D = error in departure $= \frac{\Sigma D}{D} \times d_{AB}$ (All +ve)

$= \frac{1.007}{540.993} \times d_{AB} = 0.181$

0.00186

[errors are +ve corrections will be -ve].

Similarly

Line BC $E_L = 0.082$ m
$E_D = 0.183$ m

Line CD $E_L = 0.786$ m
$E_D = ~~0.183~~$ 0.139 m

Line DE $E_L = 0.016$ m
$E_D = 0.194$ m

Line EA $E_L = 0.541$ m
$E_D = 0.307$ m

(2)

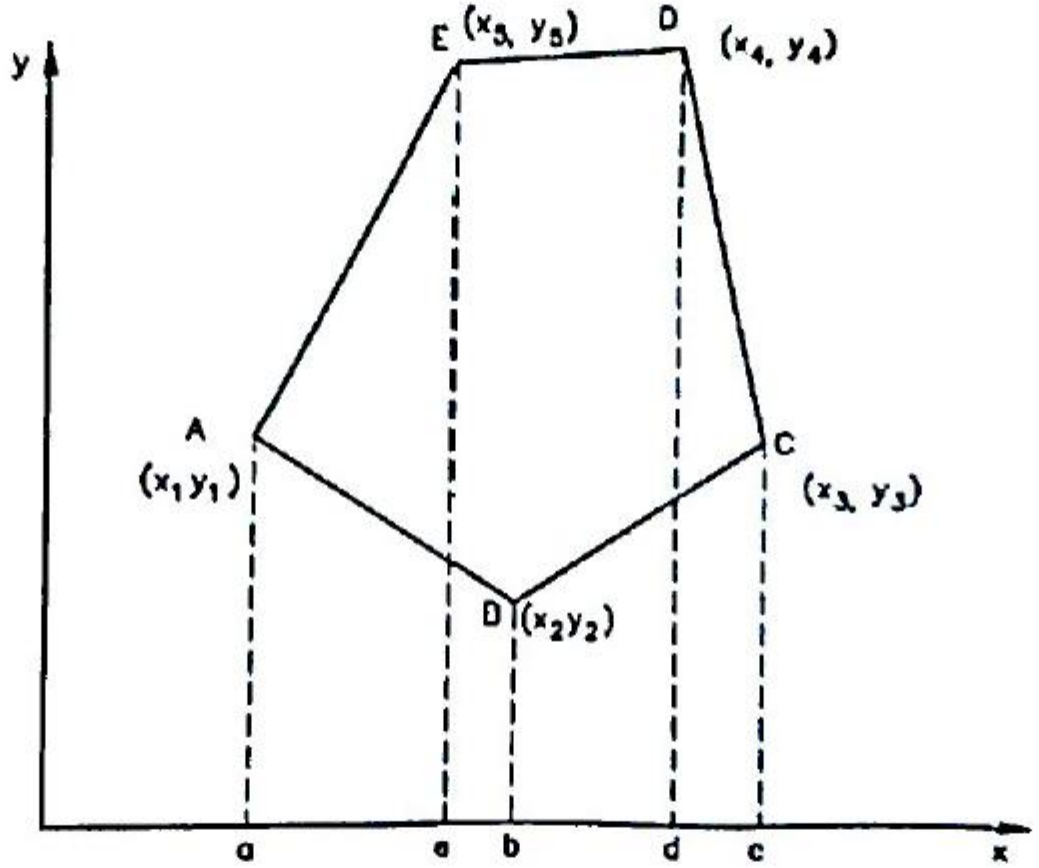
Line	Correct Latitude	Correct departure
AB	-70.638	97.483
		98.416
BC	17.185	74.598
CD	163.843	-104.785
DE	3.484	-165.709
EA	-113.865	
	ΣL = 0.009	ΣD = 0.003

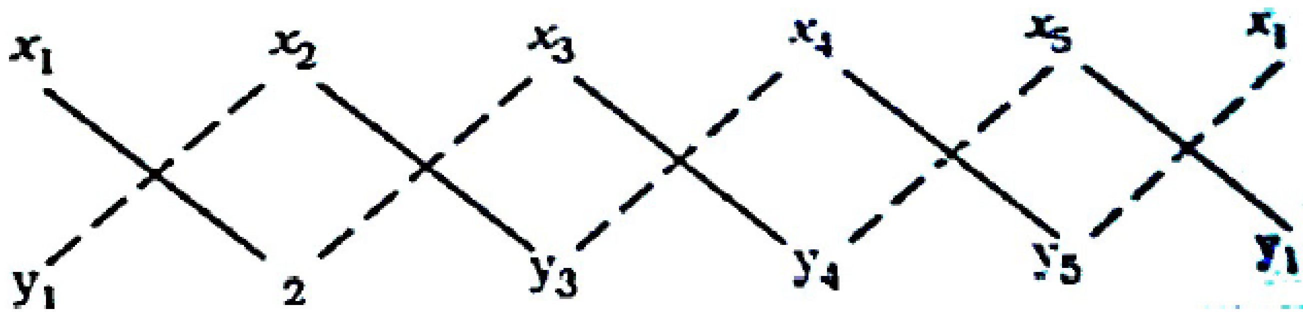
ok

Calculation of Area of traverse

- Co-ordinate method
- Meridian Distance method
- Double Meridian Distance method (DMD)
- Double parallel distance method
- Departure and total Latitude method

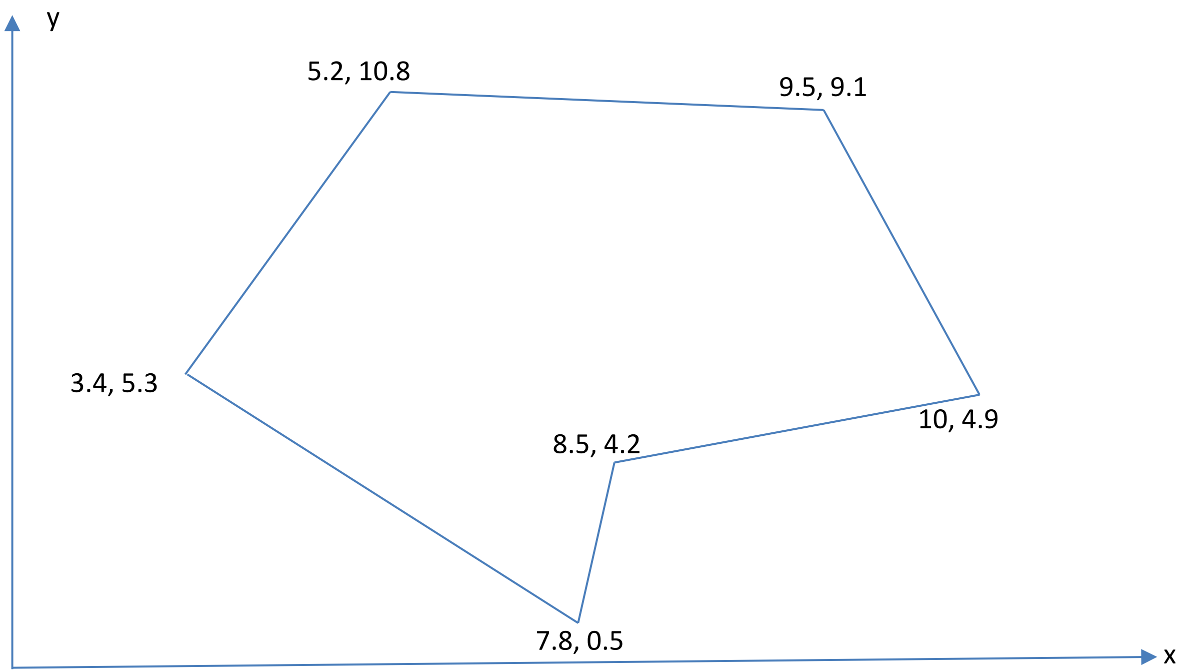
Co-ordinate method





- $A = \frac{1}{2}\left[(x_1y_2) + (x_2y_3) + (x_3y_4) + (x_4y_5) + (x_5y_1) - (x_1y_5) - (x_5y_4) - (x_4y_3) - (x_3y_2) - (x_2y_1)\right]$

Q3: Find the area of the following figure. The co-ordinates of the points are given in m.



Point	x-coordinate	y-coordinate
A	3.4	5.3
B	7.8	0.5
C	8.5	4.2
D	10	4.9
E	9.5	9.1
F	5.2	10.8
A	3.4	5.3

$$\text{Area} = \frac{1}{2}[(3.4 \times 0.5 + 7.8 \times 4.2 + 8.5 \times 4.9 + 10 \times 9.1 + 9.5 \times 10.8 + 5.2 \times 5.3) - (5.3 \times 7.8 + 0.5 \times 8.5 + 4.2 \times 10 + 4.9 \times 9.5 + 9.1 \times 5.2 + 10.8 \times 3.4)] = 39.545 m^2$$


Thank You